

Function Composition

Create a function named ``compose`` that:

- Accepts any number of functions as arguments
- Returns a new function representing their composition
- Passes arguments to the first function
- Passes each function's output to the next function
- Returns the final output

The provided functions are:

- ``add(*args)``: Returns the sum of all arguments
- ``square(a)``: Returns the square of a number
- ``splitter(a)``: Divides a number by 2 and returns ``[(a + 1) // 2, (a + 1) // 2]``
- ``my_max(a)``: Returns the maximum value in a number or list
- ``my_min(a)``: Returns the minimum value in a number or list

****Note:**** Only the first function handles multiple arguments; subsequent functions take a single argument.

Example

```
`functionsList = [add, splitter]`
```

The function call ``compose(functionsList[1], functionsList[2])`` should return a function, call it ``composedFunctions``. If ``argumentsList = [2, 3]``, ``composedFunctions(2, 3)``, should return ``[3, 3]``.

- ``add(2, 3) -> 5``
- ``splitter(5) -> [3, 3]``

Function Description

Complete the function ``compose`` in the editor with the following parameters:

- ``*functionsList``: function arguments

Returns

- ``function``: the composition of ``functionsList``

Constraints

- $1 \leq n \leq 15$
- ``functionsList[0]`` = `add`
- ``functionsList[i]`` = {`square`, `splitter`, `my_max`, `my_min`}, where $i \neq 0$
- $1 \leq q \leq 10$, length of ``argumentsList``
- $1 \leq \text{`argumentsList[i]`} \leq 10$
- It is guaranteed that the list of functions generates a valid composition.

Input Format For Custom Testing

The first line contains an integer, n , denoting the number of elements in `functionsList``.

Each line i of the n subsequent lines (where $0 \leq i < n$) contains a string describing `functionsList[i]``.

The first line contains an integer, q , denoting the number of elements in `argumentsList``.

Each line i of the q subsequent lines (where $0 \leq i < q$) contains an integer describing `argumentsList[i]``.

Sample Case 0

Sample Input For Custom Testing

STDIN	Function
-----	-----
3	functionList[] size n = 3
add	functionList = ["add", "splitter", "my_max"]
splitter	
my_max	
3	argumentsList[] size q = 3
2	argumentsList = [2, 1, 3]
1	
3	

Sample Output

3

Explanation

``compose(2, 1, 3)`` is processed as shown below -

- ``add(2, 1, 3)`` -> 6
- ``splitter(6)`` -> [3, 3]
- ``my_max([3, 3])`` -> 3

Sample Case 1

Sample Input For Custom Testing

STDIN	Function
-----	-----
3	functionList[] size n = 3
add	functionList = ["add", "square", "splitter"]
square	
splitter	
2	argumentsList[] size q = 2
2	argumentsList = [2, 4]
4	

Sample Output

```
[18, 18]
```

Explanation

`compose(2, 4)` is processed as shown below -

- `add(2, 4)` -> 6
- `square(6)` -> 36
- `splitter(36)` -> [18, 18]

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